LU Decomposition

Procedure:

1. Row-reduce A to upper triangular U using *only* row addition (no swaps/scaling). Record: "Add $c\times$ row i to row j" $\Rightarrow$ place $-c$ at L_{ji}.
2. Elementary matrices: step "add $c\times$ row i to row j" = matrix E (identity with c at (j,i)). After all steps: $E_s \cdots E_1 A = U$.
3. $L = E_1^{-1} E_2^{-1} \cdots E_s^{-1}$. Inverse of "add c" is "subtract c". L is lower triangular, 1s on diagonal, **negated multipliers** below.

Solving $Ax = b$ via LU: (1) Forward sub: solve $Ly = b$ top-to-bottom. (2) Back sub: solve $Ux = y$ bottom-to-top. **Cost:** Factor $O(n^3)$; each solve $O(n^2)$.

Connections between decompositions

$A = LU = QR = CR = V\Lambda V^{-1}$ (if diag.) $= U_s \Sigma V_s^\top$ (SVD).

- LU**:** Encodes row reduction. Best for solving $Ax = b$ (esp. multiple b's). $\det(A) = \prod u_{ii}$.
- CR**:** C = pivot cols of A, R = rows of RREF. Reveals rank, bases for Col/Row spaces.
- QR**:** Q orthonormal, R upper tri. Best for projections, least squares ($A^\top Ax = A^\top b$ becomes $Rx = Q^\top b$), numerical stability. Built via Gram–Schmidt.
- $V\Lambda V^{-1}$**:** Eigendecomposition. Powers A^k, matrix functions $f(A)$, stability. Square + diagonalizable only.
- $U\Sigma V^\top$ (SVD): Universal (all matrices). Fundamental subspaces, condition number, low-rank approx.

Least squares shortcut via QR: If $A = QR$ (full col rank), normal eqs $A^\top Ax = A^\top b$ simplify: $R^\top Q^\top QRx = R^\top Q^\top b \;\Rightarrow\; R^\top Rx = R^\top Q^\top b \;\Rightarrow\; Rx = Q^\top b$. So: compute $Q^\top b$, then back-substitute with R.

RREF & Fundamental Subspaces

Row reduce A to RREF R. **Rank** $r = \#$ pivots. **Nullity** $= n - r$.

Pivot columns of original $A \to$ basis for Col(A). **Nullspace basis:** For each free variable, set it to 1 (others 0), back-solve from $Rx = 0$. **Full solution to $Ax = b$:** $x = x^* + c_1 n_1 + c_2 n_2 + \cdots$ where x^* is any particular solution and $\{n_i\}$ is a nullspace basis. **Four subspaces:** Col(A) $\subseteq \mathbb{R}^m$ dim r; Null(A) $\subseteq \mathbb{R}^n$ dim $n-r$; Row(A) $\subseteq \mathbb{R}^n$ dim r; Null($A^\top$) $\subseteq \mathbb{R}^m$ dim $m-r$.

Determinants

$|\det(A)|$ = volume scaling factor of $T(x) = Ax$.

Key properties: $\det(AB) = \det(A)\det(B)$, $\det(A^\top) = \det(A)$, $\det(A^{-1}) = 1/\det(A)$. (proof: $A^{-1}A = I \Rightarrow \det(A^{-1})\det(A) = \det(I) = 1$)

Dependent columns $\Rightarrow \det(A) = 0$ (volume collapses). Row swap flips sign; scale row by c multiplies det by c; row addition preserves det.

Via LU: $\det(A) = \det(L)\det(U) = \prod_i u_{ii}$. **Via eigenvalues:** $\det(A) = \prod_i \lambda_i$.

Char. poly: $\det(A - \lambda I) = 0$. For 2×2: $\lambda^2 - \mathrm{tr}(A)\lambda + \det(A) = 0$. For upper/lower triangular or diagonal: eigenvalues are the diagonal entries. (trace = sum of the main diagonal)

Jacobian: $J_{ij} = \partial f_i/\partial y_j$. Change of var: $dx = |\det(J)|\,dy$. Polar: $\det(J) = r$.

Geometric interpretation: For square $A \in \mathbb{R}^{n\times n}$, $|\det(A)|$ is the factor by which $T(x) = Ax$ scales n-dimensional volume. In 2D: area scaling. In 3D: volume scaling. In general: hypervolume scaling. The **sign** of $\det(A)$ tells whether orientation is preserved (+) or reversed (−).

Why only square matrices: The determinant measures how a map $\mathbb{R}^n \to \mathbb{R}^n$ scales n-dimensional volume. A non-square $m \times n$ matrix maps $\mathbb{R}^n \to \mathbb{R}^m$, changing dimension, so there is no single scalar describing n-volume scaling on the whole space.

Dependent columns $\Rightarrow \det(A) = 0$: the columns span a subspace of dimension $< n$, so the image of T has zero n-dimensional volume (everything collapses to a lower-dimensional set).

Computing: Via LU: $\det(A) = \prod_i u_{ii}$. Via eigenvalues: $\det(A) = \prod_i \lambda_i$. Via cofactor expansion along row/col (small matrices).

Eigenvalues & Eigenvectors

$Av = \lambda v$, $v \neq 0$. Only square matrices (need Av and λv in same space).

Steps: (1) Solve $\det(A - \lambda I) = 0$ for λ. (2) For each λ, solve $(A - \lambda I)v = 0$ (nullspace).

Algebraic mult = times λ appears as root. **Geometric mult** = nullity$(A - \lambda I) = \#$ independent eigenvectors. Always geo $\leq$ alg.

- **Reflection** $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$: $\lambda = 1$ (unchanged), $\lambda = -1$ (reflected).
- **Rotation** $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$: $\lambda = e^{\pm i\theta} = \cos\theta \pm i\sin\theta$.
- **Projector** $P^2 = P$: $P^2 v = Pv \Rightarrow \lambda^2 v = \lambda v \Rightarrow \lambda^2 = \lambda \Rightarrow \lambda \in \{0, 1\}$.

Eigenvalue Decomposition, EVD

$A = V\Lambda V^{-1}$ iff A has n lin. indep. eigenvectors (geo=alg for all λ).
$V = [v_1 | \cdots | v_n]$, $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$.

Geometric meaning of Ax: (1) V^{-1}: change to eigenbasis. (2) Λ: scale each coord by λ_i. (3) V: return to standard basis.

Powers: $A^k = V\Lambda^k V^{-1}$. If $|\lambda| > 1$: $|\lambda^k| \to \infty$; $|\lambda| < 1$: $|\lambda^k| \to 0$.

Functions: If $f(x) = \sum \alpha_k x^k$, then $f(A) = Vf(\Lambda)V^{-1}$,
$f(\Lambda) = \mathrm{diag}(f(\lambda_1), \ldots, f(\lambda_n))$.
$A^{1/2} = V\Lambda^{1/2}V^{-1}$; $\quad e^A = Ve^\Lambda V^{-1}$; $\quad A^0 = I$.

Power Method (find dominant eigenvalue):

1. Pick random y_0. Iterate: $y_k = Ay_{k-1}$, normalize $y_k \leftarrow y_k/\|y_k\|$.
2. Converges to eigenvector of largest $|\lambda|$.
3. Eigenvalue: Rayleigh quotient $R(A, v) = \frac{v^\top Av}{\|v\|^2}$. If v is eigenvector: $R = \lambda$.

Proof $R(A, v) = \lambda$: $v^\top Av = v^\top \lambda v = \lambda\|v\|^2$, so $R = \lambda$. *Inverse power:* multiply by $(A - \mu I)^{-1}$ each step $\to$ converges to eigenvector of eigenvalue closest to μ.

Projection

projection of w onto v: $\mathrm{proj}_v w = \frac{v^\top w}{v^\top v}\, v$, Perpendicular part: $w_\perp = w - \mathrm{proj}_v w$.

Projector matrix: $P = \frac{vv^\top}{v^\top v}$. Complement: $P_\perp = I - P$.

Onto a subspace (range of A): With orthonormal Q spanning Col(A): $P = QQ^\top$, $P_\perp = I - QQ^\top$. General: $P = A(A^\top A)^{-1}A^\top$. Decompose b: $b_\parallel = Pb$ (on range), $b_\perp = (I - P)b$ (on null of $A^\top$).

Projector properties: **Projector:** $P^2 = P$. Orthogonal projector: $P^2 = P$ and $P^\top = P$. A matrix P is a projection matrix iff it is idempotent; A matrix P is an orthogonal projector iff it is both idempotent and symmetric

Kaczmarz / Stochastic Gradient Descent

Update:

$$x^{(k+1)} = x^{(k)} - \frac{a_i^\top x^{(k)} - b_i}{\|a_i\|^2}\, a_i.$$

Each step projects error perpendicular to row a_i, solving $a_i^\top x = b_i$. Error relation: $\|x^{(k+1)} - x^*\|^2 = \sin^2\theta\, \|x^{(k)} - x^*\|^2$, so error never increases.

Gram–Schmidt & QR Factorization

QR decomposition: $A = QR$. If A is $m \times n$ with rank r: Q is $m \times r$ with orthonormal columns ($Q^\top Q = I_r$), R is $r \times n$ upper triangular with positive diagonal entries.

Full Gram–Schmidt procedure on columns $a_1, \ldots, a_n$ of A:

Step 1: $r_{11} = \|a_1\|$. $\quad q_1 = a_1/r_{11}$.

Step k (for $k = 2, 3, \ldots, n$):

(a) Compute projection coefficients: $r_{jk} = q_j^\top a_k$ for each $j = 1, \ldots, k-1$.
(b) Subtract all projections:
$\tilde{q}_k = a_k - \sum_{j=1}^{k-1} r_{jk}\, q_j = a_k - (q_1^\top a_k)q_1 - (q_2^\top a_k)q_2 - \cdots$
(c) **If** $\tilde{q}_k = 0$: column a_k is a linear combination of $a_1, \ldots, a_{k-1}$ (linearly dependent). GS skips this column—do not produce a new q, set $r_{kk} = 0$. This means rank $< n$.
(d) **If** $\tilde{q}_k \neq 0$: normalize: $r_{kk} = \|\tilde{q}_k\|$, $\quad q_k = \tilde{q}_k/r_{kk}$.

Reading R entries: The entry r_{jk} ($j < k$) equals the projection of a_k onto q_j: $r_{jk} = q_j^\top a_k$. The diagonal $r_{kk} = \|\tilde{q}_k\|$ (length of orthogonal residual). Equivalently, column k of A reconstructs as: $a_k = r_{1k}\, q_1 + r_{2k}\, q_2 + \cdots + r_{kk}\, q_k$.

Projection of a new vector a onto existing $q_1, \ldots, q_k$:

$$\mathrm{proj} = (q_1^\top a)\, q_1 + (q_2^\top a)\, q_2 + \cdots + (q_k^\top a)\, q_k = QQ^\top a$$

Orthogonal residual: $a - \mathrm{proj}$.

Concrete example pattern (Practice Final Q4 style): Given q_1, q_2 and a_3: (1) $r_{13} = q_1^\top a_3$, $r_{23} = q_2^\top a_3$. (2) Projection of a_3 onto $\{q_1, q_2\}$: $p = r_{13}\, q_1 + r_{23}\, q_2$. (3) $\tilde{q}_3 = a_3 - r_{13}\, q_1 - r_{23}\, q_2$. (4) $r_{33} = \|\tilde{q}_3\|$, $\quad q_3 = \tilde{q}_3/r_{33}$. (5) Last column of R is $[r_{13},\ r_{23},\ r_{33}]^\top$.

Solving for coordinates / linear combinations of Q columns: Find c such that $Qc = x$: since $Q^\top Q = I$, simply $c = Q^\top x$, i.e. $c_j = q_j^\top x$ for each j. Then $x = c_1 q_1 + c_2 q_2 + \cdots + c_r q_r$. (Works only if $x \in$ Col(Q); otherwise Qc gives the projection of x.)

Building projectors from Q: See Projection section.

Least Squares (Comprehensive)

Problem: $Ax = b$ has no exact solution ($b \notin$ Col(A)). Find x^* minimizing $\|Ax - b\|^2$.

Method 1: Normal Equations (direct)

1. Form $A^\top A$ (size $n \times n$) and $A^\top b$ (size $n \times 1$).
2. Solve the square system $A^\top A\, x^* = A^\top b$.

Method 2: Projection then Solve

1. Find orthonormal Q for Col(A) via Gram–Schmidt.
2. Project: $b_\parallel = QQ^\top b$ (this is Ax^*, the closest point in Col(A) to b).
3. Solve the *consistent* system $Ax^* = b_\parallel$ (by reduction or back-sub).

Both methods yield the same x^*. Residual $r = Ax^* - b = b_\parallel - b$ satisfies $A^\top r = 0$ (orthogonal to Col(A)).

Calculus Derivation (HW6 Q2 style)

Let $f(x) = \|Ax - b\|^2 = (Ax - b)^\top (Ax - b)$.

Step 1: Expand. $f(x) = x^\top A^\top Ax - 2x^\top A^\top b + b^\top b$.

(*Use:* $(Ax - b)^\top = x^\top A^\top - b^\top$; distribute; note $(x^\top A^\top b)^\top = b^\top Ax$ is a scalar so equals itself transposed, giving $2x^\top A^\top b$.)

Step 2: Gradient of linear term. $\nabla_x(v^\top x) = v$.

(*Proof:* $v^\top x = \sum_k v_k x_k$, so $\partial_{x_k}(v^\top x) = v_k$, thus gradient $= v$.)

Step 3: Gradient of quadratic form. $\nabla_x(x^\top Mx) = 2Mx$ for symmetric M.

(*Proof:* $x^\top Mx = \sum_{i,j} m_{ij} x_i x_j$.
$\partial_{x_k} = \sum_j m_{kj} x_j + \sum_i m_{ik} x_i = [Mx]_k + [M^\top x]_k = 2[Mx]_k$.)

Step 4: Combine. $\nabla f = 2A^\top Ax - 2A^\top b = 0 \;\Rightarrow\; A^\top Ax^* = A^\top b$. ✓

Regression Setup (HW7 Q1 style)

Model: $g(x, \theta) = \theta_1 g_1(x) + \theta_2 g_2(x) + \cdots + \theta_n g_n(x)$.

Form A: $A_{ij} = g_j(x_i)$. Dimensions: m rows (data points) $\times$ n cols (parameters).

Linear fit: $g(x, \theta) = \theta_1 + \theta_2 x \;\Rightarrow\; A = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_m \end{bmatrix}$.

Quadratic fit: $g = \theta_1 + \theta_2 x + \theta_3 x^2 \;\Rightarrow\; A = \begin{bmatrix} 1 & x_1 & x_1^2 \\ \vdots & \vdots & \vdots \\ 1 & x_m & x_m^2 \end{bmatrix}$.

If $m > n$ and A has full column rank ($r = n$): system is overdetermined, no exact solution, use least squares.

Unweighted: Solve $A^\top A\, \theta^* = A^\top y$.

Weighted (errors σ_j differ): Let $\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_m)$. Replace $A \to \tilde{A} = \Sigma^{-1}A$, $y \to \tilde{y} = \Sigma^{-1}y$. Solve $\tilde{A}^\top \tilde{A}\, \theta^* = \tilde{A}^\top \tilde{y}$.

Key: If all $\sigma_j^2 = \sigma^2$ (constant), the σ^{-2} factors cancel and θ^* is same as unweighted. Non-constant σ_j: matters! Measurements with small σ_j get more weight (we trust them more).

Chi-squared: $\chi^2 = \|\Sigma^{-1}(A\theta^* - y)\|^2 = \sum_j \frac{(A\theta^* - y)_j^2}{\sigma_j^2}$. Measures goodness of fit.

SVD: Singular Value Decomposition

Definition & Properties

$A = U\Sigma V^\top$ for **any** $m \times n$ matrix A.

U: $m \times m$ orthonormal ($U^\top U = UU^\top = I_m$). Columns = **left singular vectors**.

V: $n \times n$ orthonormal ($V^\top V = VV^\top = I_n$). Columns = **right singular vectors**.

Σ: $m \times n$ diagonal, entries $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$ = **singular values**. Exactly r nonzero σ's ($r = \text{rank}(A)$).

Existence: SVD exists for *every* matrix—any shape, any rank, unlike eigendecomposition.

Computing the SVD (HW8 Q6 style)

Step 1: Compute $AA^\top$ ($m \times m$) and/or $A^\top A$ ($n \times n$).

Step 2: Key relations: $AA^\top = U(\Sigma\Sigma^\top)U^\top$ and $A^\top A = V(\Sigma^\top\Sigma)V^\top$.

(Proof: $AA^\top = U\Sigma V^\top V\Sigma^\top U^\top = U\Sigma\Sigma^\top U^\top$ since $V^\top V = I$.)

Step 3: Find singular values. Compute eigenvalues of $AA^\top$ (or $A^\top A$—nonzero eigenvalues are the same). Singular values: $\sigma_i = \sqrt{\lambda_i}$. Order: $\sigma_1 \geq \sigma_2 \geq \cdots$.

Step 4: Find U. Columns of U = normalized eigenvectors of $AA^\top$. Order columns so u_i corresponds to σ_i^2. Verify $U^\top U = I$.

Step 5: Find V. Columns of V = normalized eigenvectors of $A^\top A$. (Or use $v_i = \frac{1}{\sigma_i} A^\top u_i$.) Verify $V^\top V = I$ and columns are mutually orthogonal/unit length.

Step 6: Build Σ. $m \times n$ matrix with σ_i on diagonal, zeros elsewhere.

Step 7: Verify $A = U\Sigma V^\top$.

Geometric Interpretation (Practice Final Q8)

$T(x) = Ax = U\Sigma V^\top x$: (1) $V^\top$: rotate/reflect in $\mathbb{R}^n$. (2) Σ: stretch axis j by σ_j (and map $\mathbb{R}^n \to \mathbb{R}^m$). (3) U: rotate/reflect in $\mathbb{R}^m$.

Unit ball → ellipsoid: The set $\{x : \|x\| = 1\}$ maps to an ellipsoid with semi-axis lengths $\sigma_1, \sigma_2, \ldots$ along directions $u_1, u_2, \ldots$.

Action on singular vectors: $Av_j = \sigma_j u_j$. Each right singular vector maps to σ_j times the corresponding left singular vector.

Using the SVD (Practice Final Q9, all parts)

Let $A = U\Sigma V^\top$ be $m \times n$, rank r with $r < m$ and $r < n$.

(1) Orthonormal bases for all four fundamental subspaces:

- $\text{Col}(A) = \text{span}\{u_1, \ldots, u_r\}$ (first r cols of U).
- $\text{Null}(A^\top) = \text{span}\{u_{r+1}, \ldots, u_m\}$ (last $m-r$ cols of U).
- $\text{Row}(A) = \text{Col}(A^\top) = \text{span}\{v_1, \ldots, v_r\}$ (first r cols of V).
- $\text{Null}(A) = \text{span}\{v_{r+1}, \ldots, v_n\}$ (last $n-r$ cols of V).

Orthonormal projector onto Col($A^\top$): $P = V_r V_r^\top$ where $V_r = [v_1 | \cdots | v_r]$.

Onto Col(A): $P = U_r U_r^\top$.

(2) SVD of related matrices:

$A^\top = V\Sigma^\top U^\top$. (Swap roles of U, V; transpose Σ. Singular values unchanged.)

A^{-1} (exists only if $m = n = r$): $A^{-1} = V\Sigma^{-1}U^\top$ where $\Sigma^{-1} = \text{diag}(1/\sigma_1, \ldots, 1/\sigma_n)$.

(3) Condition number: $\kappa(A) = \sigma_1/\sigma_r = \sigma_{\max}/\sigma_{\min}$ (ratio of largest to smallest *nonzero* singular value). Large κ = ill-conditioned (small input perturbation → large output change).

(4) Outer product expansion & low-rank approximation:

$A = \sum_{i=1}^r \sigma_i\, u_i v_i^\top$ (sum of r rank-one matrices).

Eckart–Young theorem: The rank-ℓ matrix B minimizing $\|A - B\|_F^2$ is:

$$B_\ell = \sum_{i=1}^{\ell} \sigma_i\, u_i v_i^\top$$

(keep only the ℓ largest singular values/vectors).

Approximation error: $\|A - B_\ell\|_F^2 = \sigma_{\ell+1}^2 + \sigma_{\ell+2}^2 + \cdots + \sigma_r^2$.

Eigenvectors & Eigenvalues: Definitions

Definition: Let A be an $n \times n$ matrix. A nonzero vector $v \in \mathbb{R}^n$ is an **eigenvector** of A with corresponding **eigenvalue** λ if $Av = \lambda v$. That is, A acts on v by simply scaling it by λ—the direction is unchanged (or reversed if $\lambda < 0$). The eigenvalue λ can be any scalar (positive, negative, zero, or complex). The eigenvector v must be nonzero.

Why rectangular matrices don't admit eigenvectors: The equation $Av = \lambda v$ requires Av and λv to live in the *same* space. If A is $m \times n$ with $m \neq n$, then $v \in \mathbb{R}^n$ but $Av \in \mathbb{R}^m$, while $\lambda v \in \mathbb{R}^n$. Since $\mathbb{R}^m \neq \mathbb{R}^n$, the equation $Av = \lambda v$ cannot hold—the left side and right side are vectors in different spaces. Eigenvectors require a map from a space to itself ($\mathbb{R}^n \to \mathbb{R}^n$), which only square matrices provide.

Finding eigenvalues: Solve $\det(A - \lambda I) = 0$ (characteristic polynomial). For 2×2: $\lambda^2 - \text{tr}(A)\lambda + \det(A) = 0$. For triangular/diagonal: eigenvalues = diagonal entries.

Finding eigenvectors: For each λ, solve $(A - \lambda I)v = 0$, i.e., find the nullspace of $(A - \lambda I)$.

Eigenvalue Decomposition (EVD)

Definition: A square matrix A is **diagonalizable** if it can be written as $A = S\Lambda S^{-1}$ where: the columns of S are eigenvectors of A; $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$ is diagonal with the corresponding eigenvalues; and S is invertible.

Necessary and sufficient condition: A has n linearly independent eigenvectors (equivalently, geometric multiplicity = algebraic multiplicity for every eigenvalue).

Sufficient conditions (either one guarantees diagonalizability):

- A has n *distinct* eigenvalues (distinct ⇒ independent eigenvectors).
- A is real symmetric ($A = A^\top$). Then A is *orthogonally* diagonalizable: $A = Q\Lambda Q^\top$ with Q orthonormal (so $S^{-1} = Q^\top$, which is easy to compute).

Checking diagonalizability (HW8 Q3 style): (1) Find char. poly, get eigenvalues and algebraic multiplicities. (2) For each eigenvalue λ: compute nullity$(A - \lambda I)$ = geometric multiplicity. (3) If geo = alg for *every* λ ⇒ diagonalizable. If any geo < alg ⇒ not diagonalizable. (4) Collect eigenvectors into S, verify S invertible (e.g., Gauss–Jordan to find S^{-1}), confirm $A = S\Lambda S^{-1}$.

Before Midterm

Existence / uniqueness / solution set

- **Existence (consistency):** $Ax = b$ is solvable ⇔ $b \in C(A)$.
- **General solution form (if consistent):** if $Ax^* = b$ for some x^*, then the set of all solutions is
$$\{x : Ax = b\} = x^* + N(A) = \{x^* + z : \ z \in N(A)\}.$$
Proof idea: if $Ax = b$ and $Ax^* = b$, then $A(x - x^*) = 0$ so $x - x^* \in N(A)$; conversely, if $z \in N(A)$ then $A(x^* + z) = Ax^* + Az = b + 0 = b$.
- **Geometry / dimension (if consistent):** $\{x : Ax = b\}$ is an *affine* subspace parallel to $N(A)$, so
$$\dim(\{x : Ax = b\}) = \dim N(A) = \text{nullity}(A) = n - r.$$
(It is a linear subspace iff $b = 0$.)
- **Uniqueness (when consistent):** the solution is unique ⇔ $N(A) = \{0\}$ ⇔ nullity$(A) = 0$ ⇔ $r = n$ (no free variables; full column rank).
- **Rank conditions (compact):**
  - $Ax = b$ is solvable for *every* $b \in \mathbb{R}^m$ ⇔ $C(A) = \mathbb{R}^m$ ⇔ $r = m$ (full row rank; pivot in each row).
  - $Ax = b$ (when solvable) has a *unique* solution ⇔ $r = n$ (full column rank; pivot in each column).
- **Wide case $m < n$:** $r \leq m < n \Rightarrow r < n \Rightarrow$ nullity$(A) = n - r > 0$, so if consistent there are infinitely many solutions; otherwise none. (Never unique.)
- **Tall case $m > n$:** $r \leq n < m \Rightarrow \dim C(A) = r < m$, so $C(A) \neq \mathbb{R}^m$ and there exist b with no solution. (Some b may still be consistent.)
- **Consistency depends on b:** rank/independence (pivots) is a property of A. Whether $Ax = b$ is consistent depends on b via the augmented system $[A \mid b]$ (e.g., a row $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$ signals inconsistency).

- Shapes: $A \in \mathbb{R}^{m \times n}$; square if $m = n$. Column/row vectors.
- Identity I_n: $AI_n = A = I_m A$ (sizes match). Upper/lower triangular.
- Matrix-vector Ax:
  - Row view: $(Ax)_i = (\text{row } i \text{ of } A) \cdot x$.
  - Column view: $Ax = x_1 a_1 + \cdots + x_n a_n$ (columns a_j of A).
- Matrix-matrix AB:
  - Entrywise: $(AB)_{ij} = \sum_k A_{ik} B_{kj}$.
  - Columnwise: $AB = [Ab_1 \ \cdots \ Ab_q]$ (columns b_j of B).
  - Rowwise: rows of AB are linear combos of rows of B.
- Dimension rule: A is $m \times n$, B is $p \times q$: AB valid iff $n = p$, then AB is $m \times q$.
- Properties: $A(B - C) = AB - AC$; $A(BC) = (AB)C$; generally $AB \neq BA$.

FTLA (dimensions + orthogonality)

- Four fundamental subspaces of $A \in \mathbb{R}^{m \times n}$: $C(A) \subseteq \mathbb{R}^m$, $N(A^\top) \subseteq \mathbb{R}^m$; $C(A^\top) \subseteq \mathbb{R}^n$, $N(A) \subseteq \mathbb{R}^n$.
- Dimensions: $\dim C(A) = \dim C(A^\top) = r$; $\dim N(A) = n - r$; $\dim N(A^\top) = m - r$.
- Orthogonality (both parts):
$$N(A) = (C(A^\top))^\perp \text{ in } \mathbb{R}^n, \qquad N(A^\top) = (C(A))^\perp \text{ in } \mathbb{R}^m.$$
- **Meaning of "orthogonal to range(A)":** $y \perp C(A)$ means $y^\top b = 0$ for *all* $b \in C(A)$. Since $b = Ax$, this becomes $y^\top Ax = 0$ for all x.
- **Proof template: $N(A^\top) = C(A)^\perp$ (both directions):**
  - ($\subseteq$) If $y \in N(A^\top)$ then $A^\top y = 0$. For any $b \in C(A)$, $b = Ax$: $y^\top b = y^\top Ax = (A^\top y)^\top x = 0$, so $y \in C(A)^\perp$.
  - ($\supseteq$) If $y \in C(A)^\perp$, then $y^\top Ax = 0$ for all x; but $y^\top Ax = (A^\top y)^\top x$. Choose $x = A^\top y$ to force $(A^\top y)^\top (A^\top y) = \|A^\top y\|^2 = 0 \Rightarrow A^\top y = 0$.
- **Practice Q6.3: null(A) closed under linear combination:** $Ax = 0, Ay = 0 \Rightarrow A(\alpha x + \beta y) = \alpha Ax + \beta Ay = 0$.
- **Consistency via FTLA (useful test):**
$$Ax = b \text{ solvable} \iff b \in C(A) \iff b \perp N(A^\top) \iff y^\top b = 0 \ \forall y \in N(A^\top).$$

Transpose rules: $(A^\top)^\top = A$; $(A + B)^\top = A^\top + B^\top$; $(cA)^\top = cA^\top$; $(AB)^\top = B^\top A^\top$; $(ABC)^\top = C^\top B^\top A^\top$.

textbfProof template for $(AB)^\top = B^\top A^\top$:

$(AB)_{ij} = \sum_k A_{ik} B_{kj} \Rightarrow (AB)_{ij}^\top = (AB)_{ji} = \sum_k A_{jk} B_{ki} = (B^\top A^\top)_{ij}$. Symmetric: $A^\top = A$. Facts: $(A^{-1})^{-1} = A$; $(AB)^{-1} = B^{-1}A^{-1}$. **Proof template for $(AB)^{-1}$:** verify $(AB)(B^{-1}A^{-1}) = I$ and $(B^{-1}A^{-1})(AB) = I$. **Gauss–Jordan method:** row-reduce $[A \mid I]$ to $[I \mid A^{-1}]$ (works iff A invertible). **When does A admit an inverse as a map $x \mapsto Ax$?** Must be bijective (one-to-one and onto). For matrices: must be square and full rank ($m = n = r$). **Equivalent conditions (square $n \times n$):**

$$A^{-1} \text{ exists} \iff r = n \iff N(A) = \{0\} \iff \text{pivots in every column}$$

$$\iff \text{columns independent} \iff \text{columns span } \mathbb{R}^n \iff Ax = b \text{ has a unique solution } \forall\, b \in \mathbb{R}$$